HIMANSHU SINGH

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SUMMARY

Backend-focused Software Engineer with 6+ years of experience designing and scaling distributed systems. Proficient in Golang, Java, and C++, with a track record of building high-performance microservices, AI-assisted / agent-driven development workflows, and end-to-end data pipelines processing 10M+ daily events. Reduced system latencies by up to 5x and delivered measurable cost savings through in-house platform development and data platform cost optimization. Experienced across EdTech, AgriTech, and Telecom domains.

SKILLS

Languages: Golang, Java, C++

Databases: MySQL, Redis, ClickHouse

Messaging / Infra: Kafka, Kubernetes, Docker, AWS

Data Platform: ClickHouse Cloud, Debezium CDC, Apache Spark (EMR), Airflow

Concepts: Distributed Systems, Microservices, BFF Architecture, API Gateway, Caching, A/B Testing, JWT Auth, RBAC

AI Tools: Claude Code, Cursor AI (AI coding agent / vibe-coding), AI-assisted & agentic development workflows

EXPERIENCE

ALLEN Digital | Bengaluru, IN

Software Engineer 2 → Senior Software Engineer *Jun 2023 – Present*

Audience Manager (In-House Backend Platform)

- Architected and built Audience Manager entirely in-house — a backend platform for audience segmentation and targeting — eliminating a costly third-party vendor dependency.
- Owned the system end-to-end (data modeling, service design, integration with downstream consumers), driving recurring annual cost savings of up to $30K through the in-house platform while giving the team full backend control over audience data.

Offline Attendance Management (RFID / Facial-Recognition Platform)

- Built an end-to-end data pipeline ingesting 10M+ daily punch and facial-recognition records from RFID and facial-recognition devices across 100+ centers pan-India.
- Designed Auto SMS, Leave Management, Holiday Management, and Absent Notification systems from scratch, secured with RBAC; built daily scheduled jobs to compute student attendance across 100+ centers.
- Vibe-coded the entire admin UI end-to-end using the Cursor AI coding agent, adhering to best practices and coding standards throughout.

ClickHouse Data Lifecycle Management (Archival)

- Designed and led an automated data lifecycle management (DLM) system for ClickHouse Cloud, archiving bronze-layer data across 50+ databases and 400+ tables to S3 with verified, checksum-based export and safe partition drop — reducing ClickHouse warehouse storage and backup costs by ~$25K annually.
- Built automated table onboarding via ClickHouse system-table discovery and schema validation, plus day-by-day archival batching orchestrated through Airflow on EMR Serverless, scaling reliably to 400+ production tables.

Data Pipelining & Analytics

- Built scalable backend data-ingestion pipelines into ClickHouse, enabling faster analytics queries and cutting reporting latency by ~40%.

Backend Platform & Experimentation

- Engineered Page Service with advanced multi-layer caching and proper cache-invalidation strategy, improving response time by 4x.
- Built a BFF service using scatter-gather architecture, reducing frontend request complexity and improving end-to-end reliability.
- Integrated an A/B Experimentation Platform supporting feature-flag rollouts to 10k+ daily active users, and developed custom Kong API Gateway plugins for request-metadata enrichment.

Mentorship & Team Enablement

- Mentored junior engineers on day-to-day tasks and task prioritization, and led solutioning brainstorms to help scope and de-risk their work.
- Conducted regular code reviews to reinforce best coding standards, and supported junior engineers' technical growth through consistent, constructive feedback.

Nurture.farm | Bengaluru, IN

Senior Software Engineer *Apr 2021 – Jun 2023*

- Developed a scalable Payment Service processing 1k+ monthly transactions via multiple PSP integrations, with 99.9% uptime.
- Built a low-latency Auth Service generating OTPs and validating JWTs with sub-50ms p99 response times.

Oracle India Pvt. Ltd. | Gurgaon, IN
Software Developer *Jul 2019 – Apr 2021*

- Optimized LTE node HSS (Home Subscriber Server) performance, achieving a 5x improvement in throughput.
- Deployed and maintained Redis caching for session management, reducing database load by 30% and improving session retrieval speed.

EDUCATION

Bachelor of Engineering — Computer Engineering
Pune University | 2015 – 2019 | GPA: 9.06 / 10

CODING PROFILES

LeetCode: leetcode.com/u/himanshu **GitHub:** github.com/himanshu-singh